

RO Systems: Compliance, SOP & Optimization Guide

Purpose:

Removes dissolved solids and impurities from water for use in sensitive processes such as pharma, labs, and food production

Why It's Ignored:

Operators may lack training on membrane care, and performance degradation is often gradual and unnoticed.

Must-Have Docs:

- Membrane replacement logs
- Water quality testing records (conductivity, TDS)
- Pretreatment and flushing SOPs
- Cleaning chemical usage logs

Compliance Officials:

FDA(21CFR Part 129 for water), EPA Safe Drinking Water Act, cGMP, ISO 14644 (cleanroom environments)

PPE Required:

Gloves, faceshield, chemical apron (for cleaning and cartridge changes)

Installation Practices:

- Pretreatment(softener, carbon, filters)
- Pressure gauges across stages
- Proper concentrate and permeate flow rates
- Accessible bypass and drain valves

Must Checks:

- Inlet/outlet pressure differential
- Water conductivity and TDS levels
- Flow rates (permeate/reject)
- System cleaning schedule adherence
- Membrane scaling or fouling signs

Regulations & Codes:

FDA 21 CFR, EPA SDWA, cGMP, USP 1231 (pharma-grade water), ISO standards